21CSE306P · APPLIED GENERATIVE AI · COURSE PROJECT 2026–27

VERITAS.

Evidence-Driven Generative AI for Reliable Research

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Problem statement

Generative AI can produce **inaccurate or unsupported information** with high confidence. Existing AI research tools often lack **claim-level verification** and **conflict detection**, making it difficult to assess the reliability of generated information.

Retrieval grounds a model in documents, but it can still add facts the evidence never states. **VERITAS catches hallucinations claim by claim** and shows how much to trust every part of the answer.

Objectives

- **O1** Ground every statement in the user's own documents
- **O2** Verify each claim individually against evidence
- **O3** Detect conflicting evidence between sources
- **O4** Give a reliable confidence score for every claim

End users

Students & researchers
literature reviews with checked, citable answers

Teachers & reviewers
see exactly where an AI answer comes from

Policy, health & legal
catch unsupported statements in critical documents

Journalists
verify claims against the original sources

Data sources

User data — PDF, DOCX, TXT or MD uploaded by the user: the **only evidence** answers may use

Demo corpus — 7 open-access arXiv papers on LLM hallucination

Benchmark — **SciFact** (Wadden et al., 2020): 5,183 abstracts, 300 expert-labelled claims

How VERITAS works: a fixed, single-pass generative pipeline

Knowledge indexing *once per document*

Upload — The user's documents on any topic → **Clean & chunk** — ~220-word passages, bibliographies removed → **Embed** — MiniLM: 384 numbers per passage → **Vector index** — Search by cosine similarity

Grounded generation *RAG*

Question — Any research question → **Retrieve** — Top-20 → cross-encoder → best 5 [S1–S5] → **Grounded prompt** — Use only the evidence, cite every sentence → **Generate** — gpt-oss-120b answer with [S#] citations

Verification & scoring *core contribution*

Claim extraction — Answer split into atomic facts → **NLI check** — DeBERTa-v3: entail, contradict or neither? → **Self-consistency** — 4 re-sampled answers: is the fact repeated? → **Verified report** — Verified · Uncertain · Unsupported · Contradicted

$$confidence = 0.6 \times NLI + 0.4 \times consistency - 0.5 \times contradiction$$

Verified ≥ 0.70 · Unsupported < 0.40 · weak claims corrected once · conflicting sources flagged

Concepts used

ML	Confidence fusion, thresholds, F1 / AUROC evaluation
DL	Neural embeddings for meaning-based search
NLP	Chunking, claim decomposition, NLI
Transformer	Bi-encoder, cross-encoder, encoder, decoder
GenAI	RAG, sampling, prompt engineering

Models used

GENERATOR · gpt-oss-120b
open-weight LLM: writes answers, extracts claims

VERIFIER · DeBERTa-v3-base NLI
trained on MNLI, FEVER & ANLI: checks claims

EMBEDDER · all-MiniLM-L6-v2
384-dimensional embeddings for search

RE-RANKER · ms-marco-MiniLM-L-6-v2
cross-encoder: selects the 5 best passages

Prompt engineering

Role-play — system persona
Instructional — grounding rules
Few-shot completion — claim extraction
Classification — LLM-as-judge baseline
Reasoning (CoT) — verdict explanation
Comparison — conflict explanation
Information — closed-book baseline
Creative — plain-language summary

Outputs & results

Generated output

"How can hallucinations in LLMs be detected?" (7 research papers)

Token-level classifiers trained on synthetic noisy data can label each token as hallucinated or not.	0.99
Hallucinatory outputs show concentrated attention and lower attention entropy.	0.96
"Generate-then-refine" pipelines can identify and correct hallucinations.	0.92
Factored verification variants improve performance. *NLI 1.00, but only 31% of re-samples repeat it*	0.72

Is the checker reliable?

DeBERTa-v3 vs. expert labels, SciFact (340 pairs)

0.70	0.69	0.86
accuracy	macro-F1	AUROC

	Supported	Contradicted	NEI
Supported	95	12	31
Contradicted	12	42	17
NEI	20	10	101

rows = experts · cols = model · NEI = not enough info

Prompt design vs. hallucination

Same evidence; only the prompt changed.

Bad prompt (vague)	74%
Zero-shot instructions	8%
Role + instructions	0%
Few-shot	17%
Chain-of-thought	0%

74% → 0% prompt engineering alone (3-question preview)

Tech stack

gpt-oss-120b · **DeBERTa-v3** · **MiniLM** · **ms-marco cross-encoder** · Python · PyTorch · Hugging Face · Sentence-Transformers · ChromaDB · Groq API · FastAPI · React

Future work

- Fine-tune the verifier on SciFact
- Learn the scoring weights from data
- Read tables and figures inside PDFs
- Complete the full RQ1, RQ3 and RQ5 study

References

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